Fiscal Year:	FY 2009	**Task Last Updated:**	FY 01/09/2009
PI Name:	Chang, Polly Y. Ph.D.		
Project Title:	Tissue-specific acute and late molecular surveillance of particle radiation effects		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	polly.chang@sri.com	**Fax:**	FY 650-859-2889
PI Organization Type:	NON-PROFIT	**Phone:**	650-859-2549
Organization Name:	SRI International		
PI Address 1:	Biosciences, PN 187		
PI Address 2:	333 Ravenswood Avenue		
PI Web Page:			
City:	Menlo Park	**State:**	CA
Zip Code:	94025-3453	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	11/01/2007	**End Date:**	10/31/2010
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX07AV20G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The space atmosphere is comprised of wide spectrum of highly energetic and charged particle radiation. Exposure to this type of radiation in the galactocosmic environment or unplanned solar particle events (SPE) may pose a potential health hazard to humans in long-term manned space explorations. A large database of information is available on the mutagenic potential of particle radiation on cultured cells, however, limited information is available for multicellular tissue responses. This is primarily due to constraints in the selection of adequate markers in the appropriate tissue systems. Transgenic rodent mutation model systems are widely accepted by regulatory agencies as an appropriate in vivo systems to access the potential carcinogenicity of a variety of test articles. In these systems, multiple copies of the inert reporter gene are integrated into every cell of the genome, making it possible to use it as a tool to measure, both quantitatively and qualitatively, mutations in every tissue of the animal. In our studies, we are using one of the transgenic mouse mutation model systems to evaluate the immediate and long-term mutagenic potential of simulated space radiation beams in different tissue. Knowledge gained in this ground-based work is directed at understanding how		

	tissue and organ respond to exposures both on the lunar surface as well as in deep space. Information gathered from this research will aid in reducing the uncertainties associated with predicting risks both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate any potential negative effects in space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The space atmosphere is comprised of particle radiation in a wide spectrum of higly energetic and charged (HZE). Exposures to these HZE radiations poses a serious health hazard to humans in lng-term manned space explorations. We are using the plasmid-based lacZ transgenic mouse mutation model system to examine the acute and long-term tissue specific mutagenic potential after exposure to radiation with different qualities. Knowledge gained in this ground-based work is directed at understanding tissue and organ response to space radiation and will aid both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate the negative effects in space flights.
Task Progress:	C57lacZ animals that are either Trp53 wild type, heterozygous or nullizygous were exposed to a range of doses of low dose-rate Solar Particle Event (SPE) simulated radiation at the NSRL at Brookhaven National Laboratory. The spectrum of proton energies range from 55 MeV/u to 150 MeV/u with about 97% of the proton dose achieved with the 55 MeV/u energy, <2% of the dose was from 75 MeV/u energy and the rest of the dose was balanced with energies between 100 to 150 MeV/u. The average dose rate used was ~1 cGy/min (or ~50 cGy/hr). Animals were exposed to total doses ranging from 50 cGy to 200 cGy. Beam uniformity measurements were performed to ensure even dose distribution across the field of exposure. Peripheral blood was harvested from animals 1, 2, 3 and 12 days after total body irradiation (TBI). We measured the changes in the erythrocyte distribution in peripheral blood in Trp53 wild type animals as a function of dose and time after SPE exposure. Within 1 day after irradiation, we noted a dramatic decrease in the total circulating reticulocytes in doses as low as 50 cGy. Higher doses at 1 and 2 Gy did not seem to result in further reduction in the total RET counts. Dose-dependent recovery was observed at 12 days post irradiation where the average % of reticulocytes/1000 erthrocytes rebounded to levels above the controls at 1 to 2 Gy. To examine the impact of allelic copy #s of Tp53 on cytogenetic damage measured in the peripheral blood, we compared the responses in %RET levels and %MN-RET obtained from 1 Gy-irradiated animals with different Tp53 genetic backgrounds. The % RET in animals that have either 1 or both functional copies of Tp53 decreased rapidly within 1 – 2 days after SPE exposure but recovery ensued after 3 days and the levels of RET appeared to be comparable to the control levels by 12 days after exposure. The % RET levels obtained from animals lacking Tp53 function were not significantly different from the controls up to 12 days post irradiation. On the other hand, the levels of %MN-RET in peripheral blood was elevated to peak levels 48 hrs after radiation exposure and appeared to be independent of Tp53 background status. Full recovery from micronucleaus induction was evident within 12 days after SPE exposure for all animals. Using the PCR Arrays specific for mouse neurotrophin and receptor genes, we quantitatively analyzed the expression of a panel of genes that have functional significance in neuronal pathways using hippocampal tissues from cohorts of animals harvested 1 week after SPE exposure. Changes in expression in several specific genes were noted after a single 1 Gy dose at 1 week after SPE exposure. Surprisingly, instead of radiation stimulating the expression of neurotrophins, the dominant effect of 1 Gy of SPEs appeared be down-regulation of these genes. In summary, we noted no adverse clinical observations in any of the animals during radiation exposure, in transit from BNL to SRI's animal facility or at any time up to 8 weeks post irradiation. Cytogenetic results show that, the level of circulating RET in Trp53 (+/+) animals was reduced, remained depressed up to 3 days but rebounded and was significantly above background levels at 12 days after irradiation. The % MN-RET in peripheral blood was temporally regulated and SPE dose-dependant. Peak %MN-RET was achieved after 1 Gy and occurred at 48 hrs after exposure. Sensitivity of %RET changes in peripheral blood appeared to be dependent on the Trp53 genotype of the animals. % RET in Trp53(+/-) animals was similar levels to the Trp53(+/+) animals. % RET in Trp53 (-/-) animals was not significantly different from those of the unirradiated controls up to 3 days after radiation exposure. The %MN-RET levels in Trp53(+/-) and Trp53(-/-) animals increased in a time course after SPE exposure and appear to parallel the time course observed for the Trp53(+/+) animals. Gene expression of hippocampus tissue at 1 week after SPE exposure showed differential expression of a number of key genes that are known to impact LTP, synaptic function, and transmission. Our results suggest that low dose rate SPE radiation impacts the hematopoietic system. However, such events are transient and may be dependent on the Trp53 genetic background of the animals. Gene expression profile in the hippocampus demonstrated that SPE induced down regulation of neurotrophic genes important to functional integrity of CNS. Further analyses of tissues collected from these experimental animals are in progress. We expect that this will provide additional mechanistic information on the impact of low dose-rate low energy protons in animal systems.
Bibliography Type:	Description: (Last Updated: 04/26/2023)
Abstracts for Journals and Proceedings	Chang PY, Bakke J, Bunin D. "DNA Damage and Processing in Central Nervous System after Heavy Ion Exposure." NASA Human Research Program Investigators' Workshop, League City, Texas. February 4-6, 2008. NASA Human Research Program Investigators' Workshop, League City, Texas. February 4-6, 2008. http://www.dsls.usra.edu/meetings/hrp2008/pdf/Radiation/1099Chang.pdf , Feb-2008
Abstracts for Journals and Proceedings	Chang PY, Bakke J, Wang A, Bunin D. "Impact of low dose-rate solar particle event radiation in vivo." 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 20 – July 2, 2008. 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 20 – July 2, 2008. , Jul-2008
Abstracts for Journals and Proceedings	Chang PY, Bunin D, Bakke J. "DNA damage and processing in brain tissue after HZE particle radiation." 54th Radiation Research Society Annual Meeting, Boston, MA, Sept 21–25, 2008. 54th Radiation Research Society Annual Meeting, Boston, MA, Sept 21–25, 2008. , Sep-2008